

Ishaan Mehta

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U.S. Citizen

Education

University of Waterloo

Bachelor of Applied Science, Computer Engineering, Co-op

Waterloo, ON

September 2026 – Present

Experience

AutoFillPDF

Founder & Engineer

Remote

February 2025 – Present

- Built and launched a document automation platform used by 70+ monthly users with \$1,000+ MRR.
- Trained a D-FINE object-detection model on CommonForms to locate form-field bounding boxes in flattened PDFs, enabling previously non-editable documents to be completed digitally.
- Developed a production pipeline combining computer vision, document parsing, and language models to identify fields, infer field meaning, and populate forms from user context.

SoCal Rehab

ML Engineer

Remote

June 2026 – Present

- Built a clinical-document automation workflow with Vertex AI and Google Apps Script to extract patient information from PDFs and draft structured notes for clinician review.
- Reduced processing time from approximately 5 hours to 15 minutes per patient across a 100+ patient workload by automating repetitive documentation and billing steps.

Technical Projects & Robotics

Ballform | Basketball shot analyzer | [GitHub](#)

Personal Project

September 2026

- Built a basketball computer vision system combining YOLO26 pose estimation and RF-DETR detection to track players, analyze shooting form, and measure shooter–defender separation.
- Optimized Apple Silicon inference with Core ML ball detection and overlapping next-frame inference with current-frame pose processing, reducing total analysis time by 52% (173s to 84s) on a 10-second clip; ball detection was $2.3\times$ faster than the CPU baseline.

FIRST Robotics Competition

Lead Electrical Engineer & Programmer, Team 2404

Pasadena, CA

2024 – 2026

- Led electrical integration and programming for a 30+ member team, wiring and debugging robot subsystems under competition time constraints.
- Developed a Python targeting algorithm using computer-vision tracking, projectile motion, and robot velocity to align shots while the robot was moving.
- Alliance place 1st at the 2026 CA District Glendale Event and 3rd at the Aerospace Valley Regional.

Electric Bike

2024 – 2025

- Built mid-drive and hub-drive e-bike conversions, integrating motors, controllers, batteries, and custom wiring.

Skills

Languages: Python, TypeScript, Java

ML / Computer Vision: PyTorch, RF-DETR, Hugging Face, Weights & Biases, model training and deployment

Robotics / Tools: Git, Linux, Arduino, Raspberry Pi, LiDAR, IMU, Google Cloud Vertex AI, PostgreSQL